

EBeam_satadrudas - Unbalanced MZIs (TE, 1550 nm)

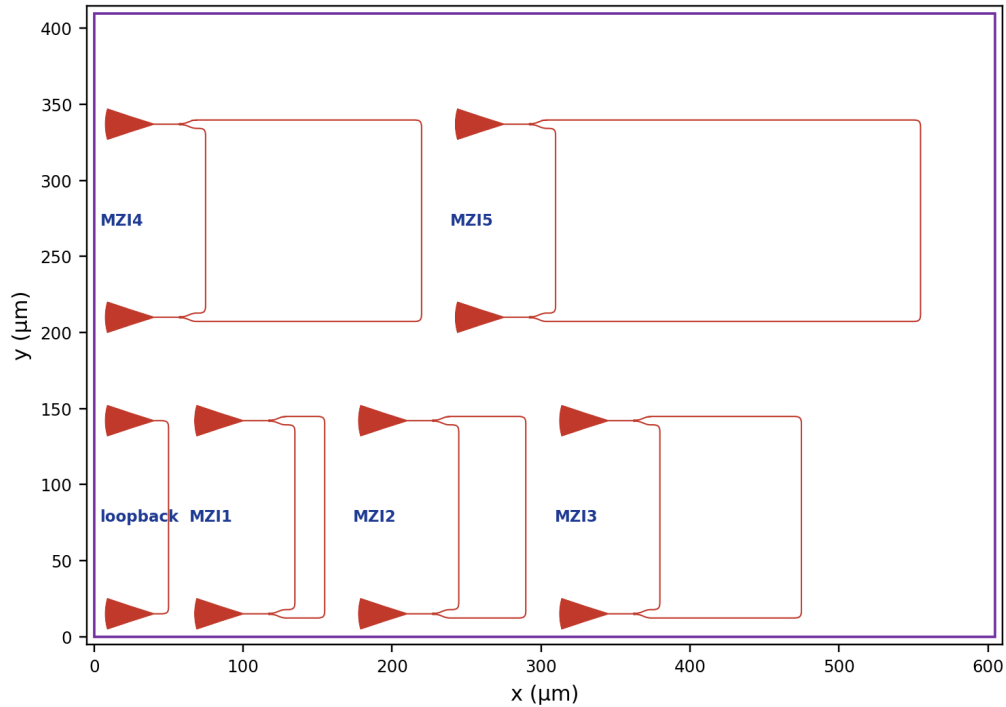
Satadru Das · Layout: submissions/EBeam_satadrudas.gds · Fork: satadrudas/openEBL-2026-10

Goal: extract the group index n_g of the 500 x 220 nm strip waveguide from the free spectral range of five Y-branch MZIs with different path-length differences, using $\text{FSR} = \lambda^2 / (n_g \Delta L)$. A grating-coupler loopback is included to remove the grating-coupler response from the measured spectra.

opt_in label	Device	ΔL (μm)	Expected FSR @ 1550 nm
opt_in_TE_1550_device_satadrudas_loopback	GC loopback	-	-
opt_in_TE_1550_device_satadrudas_MZI1	Y-branch MZI	51	11.26 nm
opt_in_TE_1550_device_satadrudas_MZI2	Y-branch MZI	101	5.69 nm
opt_in_TE_1550_device_satadrudas_MZI3	Y-branch MZI	201	2.86 nm
opt_in_TE_1550_device_satadrudas_MZI4	Y-branch MZI	301	1.91 nm
opt_in_TE_1550_device_satadrudas_MZI5	Y-branch MZI	501	1.15 nm

Simulation: a finite-element mode solver (femwell, with Si and SiO₂ dispersion) gives $n_{\text{eff}} = 2.446$ and $n_g = 4.184$ at 1550 nm. Across width 500 ± 20 nm and thickness 220 ± 5 nm, $n_g = 4.146$ - 4.224 , i.e. about $\pm 1\%$ change in FSR.

Layout: SiEPIC-EBeam-PDK cells ebeam_gc_te1550 (127 μm pitch) and ebeam_y_1550; waveguides Strip TE 1550 nm, $w=500$ nm. Design area 605 x 410 μm . Passes SiEPIC functional verification and the openEBL submission checks with 0 errors.



Layout of EBeam_satadrudas. Bottom row: loopback, MZI1-MZI3; top row: MZI4, MZI5.